Foad Kheirabady

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Research Interests

Broadly interested in AI research across several directions I haven't yet settled between: **Trustworthy AI** (robustness, safety, and reliability of real, deployed systems; practical attack and defense evaluation), **Computer Vision**, **AI for Healthcare** (human-AI collaboration, clinical decision support), and **Bioinformatics** (ML applied to biological and genomic data). Have completed coursework in **Artificial Intelligence**, and currently taking **Machine Learning** and **Data & Network Security**.

Education

- **Sharif University of Technology** — Tehran, Iran
  B.Sc. in Computer Engineering — *Sept. 2023 – Expected: June 2028*
  - **Relevant Coursework**: **Artificial Intelligence** (20/20), **Machine Learning** (in progress), **Data & Network Security** (in progress), **Theory of Machines & Languages** (Automata Theory) (17.2/20), **Design & Analysis of Algorithms** (19.5/20), Data Structures (20/20), Fundamentals of Programming (20/20), Advanced Programming (20/20), Discrete Structures (18.4/20)

Awards and Honors

- **National University Entrance Exam (Konkur)**: Ranked **28th** nationwide among over 150,000 candidates in the Mathematics and Physics track (2023).

Experience

- **Chavosh (Yektanet)** — Tehran, Iran
  Software Engineer (Part-time) — *Mar. 2025 – Aug. 2026*
  - **Backend systems**: Built and optimized web applications and RESTful APIs using **Django** and **PostgreSQL**, improving query performance and ensuring data integrity across services.
  - **Automated messaging workflows**: Developed and maintained Telegram bots with **Python** to automate internal workflows and integrate backend services, reducing manual intervention time significantly.
  - **Data processing pipelines**: Designed CSV/JSON import-export pipelines to streamline data ingestion and analysis, cutting data processing overhead for reporting tasks.
- **Sharif University of Technology** — Tehran, Iran
  Teaching Assistant – Computer Engineering Courses — *Sept. 2024 – Present*
  - **Automata Theory – Dr. Izady**: Assisted students with formal languages, finite automata, pushdown automata, and Turing machines; held office hours and prepared exercise sets.
  - **Artificial Intelligence – Dr. Tan Ghatari & Dr. Soleymani**: Assisted students with search, knowledge representation, and machine learning fundamentals across two terms; held office hours and graded assignments.
  - **Computer Networks – Dr. Dolati**: Assisted students with networking protocols, socket programming, and course projects; held office hours and graded assignments.
  - **Fundamentals of Programming (C) – Prof. Fazli**: Guided students through C syntax, debugging, and assignments; provided individualized support during lab sessions.
  - **Advanced Programming (Java) – Prof. Fazli**: Assisted students with OOP principles, project design, version control (Git), and socket-based client–server communication.
  - **Data Structures – Prof. Abam**: Supported students in implementing and analyzing core data structures (lists, trees, graphs, hash tables), with emphasis on algorithmic complexity.

Projects

- **InjectLab** (Ongoing):
  - Built a **Python** research testbed measuring indirect **prompt injection** attacks against tool-using **LLM agents** (simulated email, web, and calendar tasks) running local open models (Qwen2.5, Llama3.1) via **Ollama**, following the AgentDojo research framework at a smaller experimental scale.
  - Measured task utility, utility-under-attack, and attack success rate via deterministic world-state checkers rather than LLM-judge evaluation; implemented and evaluated defenses including spotlighting, sandwiching, tool filtering, and a classifier-based injection detector.
- **SoroushClient** (PyPI: `soroushclient`):
  - Built an asynchronous **MTProto** client library for the Soroush Plus messaging platform in pure **Python**, published to PyPI (`pip install soroushclient`).
  - Implemented a **WebSocket**-based transport layer with automatic reconnect, phone-number OTP authentication, and persistent session storage.
- **SoundWave – Full-Stack Music Streaming Platform**:
  - Developed a Spotify-like streaming service in a three-person team (**Next.js 14** + TypeScript frontend, **Django REST Framework** backend, **PostgreSQL**).
  - Owned the `accounts`, `billing`, `support`, and `notifications` apps: JWT authentication, role-based permissions, and payment-gateway integration; containerised the full stack with **Docker Compose**.
- **BeatStream – Spotify-like MVP in Go**:
  - Designed a modular **Go** backend (chi, sqlc, pgx) with clean service boundaries for Auth, Catalog, Playlist, Streaming, and Analytics.
  - Added **Redis** caching and Meilisearch indexing; automated CI/CD with Docker Compose and schema migrations.
- **NeoGit**:
  - Built a **C**-based CLI tool supporting `init`, `commit`, `log`, and `branch` commands to simulate core Git workflows, with robust memory management.

Programming Skills

- **Languages:**
  - **Python** — *Advanced*
  - **Java** — *Advanced*
  - Go — *Intermediate*
  - C / C++ — *Intermediate*
  - SQL — *Intermediate*
- **Databases:**
  - **PostgreSQL** — *Advanced*
  - Redis — *Intermediate*
- **Tools & Platforms:**
  - Git / GitHub / GitLab — *Advanced*
  - Ollama — *Intermediate*
  - Docker — *Intermediate*
- **Frameworks & Libraries:**
  - **Django & DRF** — *Advanced*
  - PyTorch — *Intermediate*
  - Hugging Face Transformers — *Intermediate*
- **Concepts:**
  RESTful APIs, OOP, MVC, Clean Architecture, Event-Driven Systems, Async Programming

Languages

- **Persian**: Native
- **English**: Advanced – professional working proficiency
- **French**: Beginner – basic conversational knowledge